

```
[root@mageshlinux downloads]# sudo rpm -qsh private-chef-11.3.2-1.el6.x86_64.rpm
warning: private-chef-11.3.2-1.el6.x86_64.rpm: Header V4 DSA/SHA1 Signature, key ID 624f920a: NOKEY
Preparing...
##### (100%)
Installing...
private-chef-11.3.2-1.el6
Thank you for installing Private Chef!
[root@mageshlinux downloads]#
[root@mageshlinux downloads]# rpm -qsh chef-server-core-12.1.2-1.el7.x86_64.rpm
warning: chef-server-core-12.1.2-1.el7.x86_64.rpm: Header V4 DSA/SHA1 Signature, key ID 624f920a: NOKEY
warning: missing file for transaction lock on /var/lib/rpm/.rpm.lock
Preparing...
##### (100%)
Updating / installing...
  1:chef-server-core-12.1.2-1.el7
##### (50%)
Cleaning up / removing...
  2:private-chef-11.3.2-1.el6
##### (100%)
```

Figure 10: Installing Chef Server components

```
[root@mageshlinux downloads]# sudo rpm -qsh chefdk-0.7.0-1.el7.x86_64.rpm
warning: chefdk-0.7.0-1.el7.x86_64.rpm: Header V4 DSA/SHA1 Signature, key ID 624f920a: NOKEY
Preparing...
##### (100%)
Updating / installing...
  1:chefdk-0.7.0-1.el7
##### (34%)
Thank you for installing Chef Development Kit!
```

Figure 11: Installing Chef Developer Kit

and enhancing it for deployment. Also, in a simulated environment, we have the flexibility to use the developer kit, debugging tools, performance tools like Nagios as well as tune the set-up before taking it to an actual environment.

In this context, we need to create a simulated environment on a VMware based Linux sandbox, and then create agent nodes using Linux VM to mimic a server and node set-up, prior to installing Chef Enterprise and Chef agents.

Setting up VMware: VMware is virtualisation software to create simulated operating environments of different flavours on a single machine. For example, we can create Linux VM in a Windows machine and work in the VM from within Windows itself. It also uses all resources like printers, the Internet, networks, etc, from the host machine in the VMs.

Using virtual environments, we can create VMs of various flavours and work on simulated environments without physically installing the operating environment. This requires the VMware workstation which can be downloaded at https://my.vmware.com/web/vmware/info/slug/desktop_end_user_computing/vmware_workstation_pro/12_0

Next, download RHEL 7 ISO and install it in the VMware workstation as shown in Figure 5.

Now select the ISO image which was just downloaded to start the installation process (Figure 6).

The next step is to validate the selections and agree to proceed with the installation (Figure 7).

Once the installation is done, install all peripheral software and tools required for Chef. Since Chef is written in Ruby and uses UNIX based tools like Curl, Mail, etc, we need to install all the required software like VMware tools, for which you need root privileges (Figure 8).

This installation process will take you through a series of steps when you run `vmware-install.pl` as instructed in Figure 8, which runs as shown in Figure 9.

You must now subscribe to Red Hat Linux to enable the subscription manager to get all downloads directly in the VM itself. If you don't subscribe to the Linux installation, any tool required for Chef installation has to be manually downloaded in the host machine as an RPM package, then ported to the

```
[root@mageshlinux epacode]# chef-server-ctl reconfigure
Starting Chef Client, version 12.5.9.current.0
Resolving cookbooks for run list: ["private-chef::default"]
Synchronizing Cookbooks:
- private-chef
- enterprise
- apt
- yum
- runit
- packagecloud
Compiling Cookbooks...
```

Figure 12: Chef configuration

```
[root@mageshlinux epacode]# chef-server-ctl test
Recipe: private-chef::cli::omsg
  * template[ntp-ntpdate.ntpdate.el6] action create (up to date)
  Recipe: private-chef::default
    * file[ntp-ntpdate-chef-server-running.json] action create (up to date)

Running handlers
Running handlers complete
Chef Client finished: 17/418 resources updated in 125.474977953 seconds
omsgcode Reconfigured!
```

Figure 13: Chef test configuration

VM, and then run manually. With the subscription option, this can be automated within the VM itself.

Now download the Chef Development Kit (DK) from <https://downloads.chef.io/chef-client/redhat/#/> and copy it to the above VM. If you have the Internet configured (proxy or direct network) in Linux VM, then use the browser in the VM (Firefox is available, by default) to download Chef Tools directly into the VM.

After installing Chef Server components (Figure 10), we can install the Chef Developer Kit (DK), which helps in writing recipes and compiling cookbooks for the Chef set-up. The screen shot in Figure 11 explains the procedure to download and install the Chef DK.

Chef configuration

Chef Server and the developer kit are now ready. Next, we need to install agents in nodes and configure them. Before starting out on this step of installing and configuring agents, let's configure the server to be fully available for use and bring it up to connect with the agents. This can be done with the *Reconfigure* option as shown in Figure 12.

We have completed the installation and configuration of Chef successfully. Now, we should validate the Chef configuration to see if it looks right and is working properly. This can be done by running a test command as shown in Figure 13. 

By: Magesh Kasthuri and Dr B. Thangaraju

Magesh Kasthuri is a senior 'Distinguished Member of the Technical Staff' in Wipro. He is a senior technical consultant in the Java Technology Practices Group and can be reached at magesh.kasthuri@wipro.com

Dr B. Thangaraju is an open source software (OSS) evangelist. He can be reached at balat.raju@wipro.com